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Climate Change and Food Security: Agricultural and Non Farm Adaptation Strategies in Asia

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Contents

Introduction	2
Methodology	4
Findings	5
The Impacts of Climate Change on Food Security	5
Adaptation in Food Production and Consumption	8
Adaptation Strategies for Food Security Management	9
Adaptation Strategies for Farm Practices	10
Recommendation	12
Conclusion	13
References	14

Abstract

Climate change (CC) poses a critical threat to global food security by intensifying food insecurity through extreme weather events, shifting climatic patterns, and altering agricultural productivity. This chapter conducts a comprehensive review of literature from 2015–2024 to assess the impacts of CC on food availability, accessibility, and utilization. It explores a range of adaptation strategies on both

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the farm level and non-farm interventions. This study highlights that CC adversely affects crop yields, exacerbates pest and disease outbreaks, and disrupts livestock and fisheries systems. The findings underscore the necessity of adopting resilient agricultural practices, such as crop diversification, conservation tillage, and enhanced water management infrastructure, to mitigate these impacts. The study examines non-agricultural adaptation strategies that promote livelihood diversification and socio-economic resilience in vulnerable populations. The chapter emphasizes the integration of adaptation, mitigation, and governance frameworks to address food security challenges. This chapter contributes to existing knowledge gaps and offers actionable recommendations for advancing food security policies and practices in the changing climate.

Keywords

Climate change · Food security · Adaptation strategies · Agricultural resilience · Sustainable practices

Introduction

Global climate change (CC) intensifies the challenge of eradicating hunger. It disrupts food systems, reduces agricultural productivity, and exacerbates food insecurity worldwide (Ullah et al. 2024). Extreme climate-related occurrences increase global food insecurity (Organization 2019; Roy et al. 2022). These severe events have become more frequent, intense, unpredictable, and persistent with the onset of CC, thereby adding to their effects across various regions (Pradhan et al. 2022). Food security is achieved when “all people, at all times, have economic, social, and physical access to sufficient, nutritious, and safe food that meets their dietary needs and food preferences for an active and healthy life” (Rinaldi 2023, p. 9). This definition of food security encompasses food availability, access, utilization, and stability (Kogan et al. 2019; Rinaldi 2023). In June 2022, nearly 345 million people faced severe food insecurity (Shyu 2023). Severe food insecurity arises when individuals or groups face acute deprivation of sufficient, safe, and nutritious food due to war, disaster, or emergencies (Brück and d’Errico 2019; Ullah et al. 2022). It creates a critical threat to lives and livelihoods, leading to starvation and disease (Organization 2022). Despite the goal of Sustainable Development Goal II to end hunger by 2030, over 670 million people will still suffer from hunger and food insecurity. Food insecurity is projected to affect 30 million people in 2030 in low-income, agriculture-dependent countries (Organization 2022). Climate extremes, population growth, and changing lifestyles intensify food insecurity in agriculture-dependent countries (Organization 2019).

Crop productivity is declining due to changes in temperature, rainfall patterns, and extreme weather events. Most research efforts have focused on determining the factors influencing food security (Mohanty et al. 2024). Existing literature examines the impact of climate extremes or CC adaptation strategies on food security (Cui and

Zhong 2024), resulting in reduced yields and increased unpredictability in agricultural outputs (Schmidt and Felsche 2024). Extreme weather conditions warm the planet and alter the crop production pattern, resulting in reduced yields and increased unpredictability in agricultural outputs. These events, such as droughts, floods, and hurricanes, are becoming more frequent and severe, directly damaging crops and disrupting planting and harvesting cycles (Harsh et al. 2024). These events can decimate local food supplies and hinder farmers' ability to recover between seasons. Variations in temperatures and precipitation patterns can change growing seasons and impact crop suitability in various regions (Naik et al. 2024). For example, staple crops such as wheat, rice, and maize have specific climate requirements, and shifts in these conditions can lead to significant declines in productivity (Dutta et al. 2024). As temperatures rise, Asia may no longer support traditional farming. It requires costly transitions to new crops or farming methods, often unfeasible for resource-poor farmers (Araya et al. 2024). The impact of CC on food availability is not confined to crop production alone. (Mohanty et al. 2024). Heat stress from higher temperatures can reduce livestock productivity, affecting milk production, reproduction rates, and overall health (Țogoe and Mincă 2024). Water scarcity, driven by CC, compounds these issues, limiting water availability for livestock and crop irrigation (Tinazzi 2024). Fisheries are similarly affected, as changing ocean temperatures and acidification alter fish distribution and reproductive cycles. For communities reliant on fishing for food and livelihood, these changes significantly threaten their food security and economic stability (Lam et al. 2020).

Food systems are interconnected and global, so climate impacts in one region can have ripple effects worldwide. For example, reduced agricultural outputs in significant exporting countries can increase global food prices, making food less affordable and accessible, especially for low-income populations (Masso 2024). The economic strain from fluctuating food prices can increase poverty and limit countries' ability to invest in sustainable agricultural practices or climate resilience measures (Chao 2024). Furthermore, regions' social and political stability can be undermined as food scarcity heightens resource competition, potentially leading to conflicts and mass migrations (Bedasa and Deksisia 2024). Adaptation and mitigation strategies are essential for addressing CC's challenges to food availability. Sustainable agricultural practices, such as crop diversification, conservation tillage, drought- and heat-resistant varieties, and integrated pest management, can enhance resilience against climate impacts (Chettri et al. 2024).

Current studies have primarily focused on CC Adaptation (CCA) strategies and their impact on food security. However, these studies lack a holistic perspective and do not describe or consider the simultaneous use of multiple adaptation strategies or strategies for sustaining agricultural activities and transitioning to alternative livelihoods. The research seeks to enhance our understanding of how food security can be improved through adaptive agricultural practices or alternative livelihoods in the context of CC. The specific objectives of this study are related to the developing countries: (1) Examine the outcomes of integrating multiple adaptation strategies on food security, acknowledging the use of a combination of approaches, and (2) Explore how adaptive agricultural practices and non-farm alternatives can enhance food security under the impacts of CC.

Methodology

This study utilized a systematic review approach (Waqas et al. 2023) of peer-reviewed literature on CC and food security presented in Fig. 1. Unlike traditional systematic reviews that seek empirical fact, the realist approach aims to explain by focusing on depth rather than extent (Denney and Tewksbury 2013). This method involves stricter inclusion criteria and qualitative critical analyses.

A keyword search was conducted in Google Scholar, ScienceDirect, IEEE, Web of Science, and ISI Web of Knowledge electronic databases using the following keywords: [nutrition OR “food security” OR “diet”] AND [“climate change” OR “global warming”]. This search aimed to ensure the research’s currency. Only articles and reviews were eligible for inclusion. The initial search results were refined by reviewing all article titles and abstracts according to the inclusion and exclusion criteria outlined in Table 1. The detailed screening and selection criteria for the articles are shown in Fig. 2.

A review strategy and four key research questions were developed to review the documents. The main research questions (RQs) addressed (1) the outcomes of integrating multiple adaptation strategies on food security, recognizing the use of a combination of approaches; (2) how adaptive agricultural practices can improve food security under the impacts of climate change; (3) non-farm options that support the sustainability of livelihoods when agricultural activities face threats from climate change; and (4) the implications of various CCA strategies on food security and suggestions for future research in this area. The realist review is well-suited for

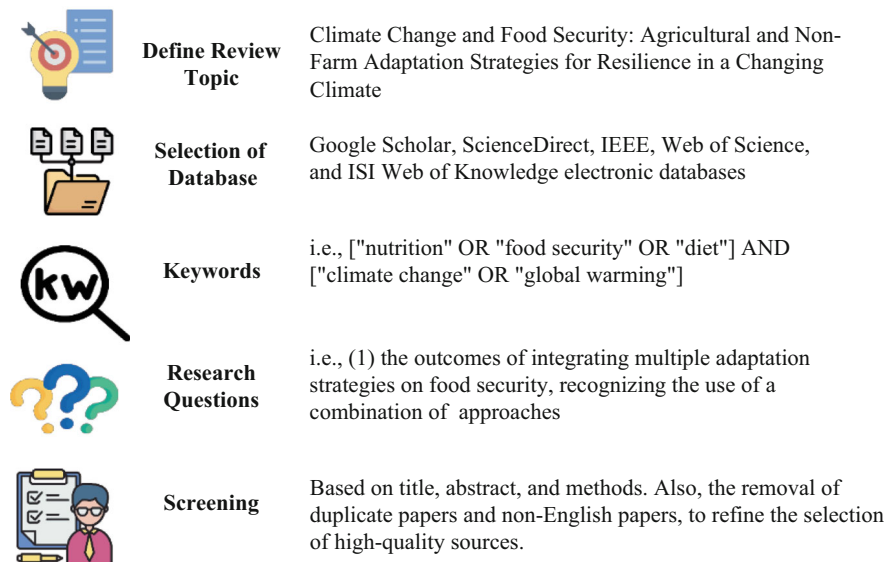


Fig. 1 Method to conduct a detailed systematic review

Table 1 Inclusion and exclusion criteria for publication selection

Included	Excluded
Published in the last ten years	CC and food losses
Articles, review articles.	Alternative food options
Adaptation for food security and its challenges	Integrating climate projections with food security analysis
CC impact on crop production	CC and the rise of infectious diseases
CC’s impact on policies	CC and food prices
CC impact on food access, availability, utilization	CC and food safety
Farmer adaptation due to CC	CC and farm techniques

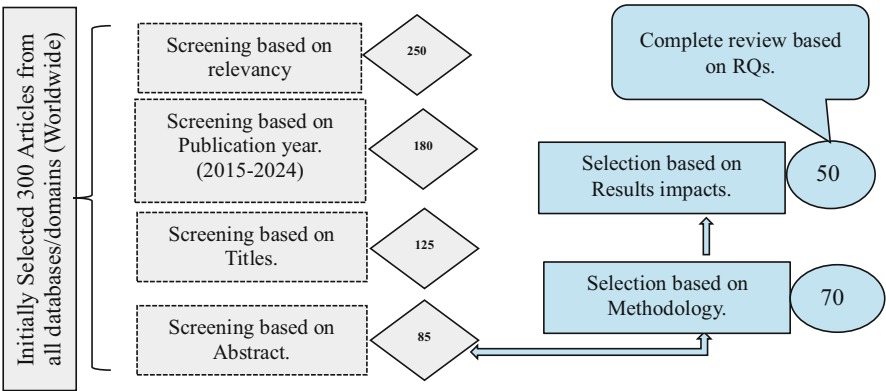


Fig. 2 Literature screening process used in the selection of articles

understanding food security as it is embedded within complex social, cultural, and ecological systems that influence vulnerability and adaptive capacity.

Findings

The Impacts of Climate Change on Food Security

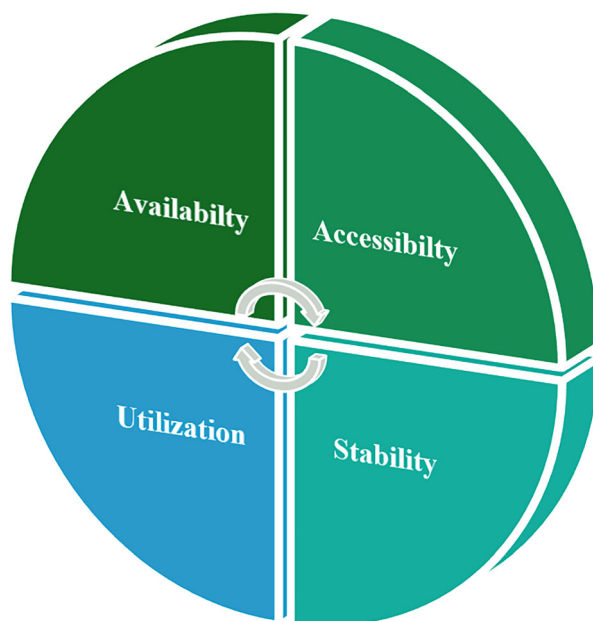
In the twenty-first century, the severity of extreme weather events, such as heatwaves, floods, droughts, typhoons, and cold waves (Lee et al. 2024; Li et al. 2024), has contributed to 70–85% of total meteorological disasters. These events have highly impacted the food production environment (Han et al. 2024). Rainfall and soil moisture are essential determinants of agricultural productivity, and the unpredictability of rainfall patterns leads to moisture instability, adversely affecting crop yields and resulting in reduced food production (Zheng et al. 2024). Furthermore, rising CO₂ levels have contributed to global temperature increases, which reduce the cultivation season, significantly impacting food production and posing a direct danger to regional food security

and yield (Lee et al. 2024). CC also alters global hydrological cycles by influencing rainfall, evaporation, and runoff, redistributing water resources on both temporal and spatial scales (Kusainova et al. 2024). Irregular rainfall patterns can reduce food production and alter food crops' nutrient composition, disrupting their inherent structural properties (Dave et al. 2024). Additionally, variations in sunshine duration negatively impact crop quality. Excessive sunlight and elevated temperatures accelerate the crop growth cycle, leading to nutrient deficiencies during the developmental stages of crop growth (Lee et al. 2024).

Food insecurity remains a significant global challenge (Tamargo et al. 2024) and is expected to intensify with increasing population growth. It is particularly severe in low- and middle-income countries (LMICs) but also affects vulnerable populations in wealthier nations (Hassen and El Bilali 2024). Food insecurity profoundly affects well-being, health, and productivity (Ferrara et al. 2024). Food insecurity requires a comprehensive approach encompassing agricultural production, CC, environmental degradation, animal and plant health, pest management, food distribution systems, and the effective allocation of scarce resources (Bedasa and Deksisa 2024). The rapid growth of the human population is a fundamental driver of food insecurity. Since the 1950s, global population growth has followed a logarithmic trend. Projections by the United Nations estimate that the global population will rise from 8 billion in 2023 to 9.7 billion by 2050, with a peak of approximately 10.4 billion in the mid-2080s, followed by an annual decline of -0.1% (van Dijk et al. 2021). The rapid growth of the human population and increased demand for resources are leading to resource depletion and the accumulation of pollutants, such as CO_2 and other greenhouse gases (GHGs), within the Earth's ecosystem, thereby contributing to CC. This combination of CC and resource depletion will challenge future food production. CC affects food access, exacerbating inequalities and creating new challenges for vulnerable populations worldwide. The main pillars of food security are shown in Fig. 3 (Guja and Bedeke 2024).

The CC is anticipated to have the most significant impact on agriculture (Ullah et al. 2024), making food prices susceptible to its adverse effects compared to non-food costs, except energy prices. Agricultural yields are highly sensitive to weather conditions and negatively affect crop yields in food-producing regions (Iliyasu and Sanusi 2024). The weather conditions result in shortages and rising commodity prices, potentially leading to global food price inflation that propagates through trade and economic networks. Numerous studies have examined the potential impact of CC on global food prices, particularly for grains. However, the Change (2007) highlights that many evaluations fail to consider the anticipated increase in the frequency of extreme weather events associated with CC. Previous and current research investigations suggest crop prices are higher when accounting for these factors than current projections indicate (Easterling et al. 2007; Nguyen et al. 2024). The influence of CC on inflation is significant in both developed and developing countries (Iliyasu et al. 2023; Iliyasu and Sanusi 2024). Various methodologies have been employed to estimate this impact, including the panel VAR model (Mukherjee and Ouattara 2021), panel local projections, fixed effects panel models (Heinen et al. 2019), and structural VAR models (Iliyasu et al. 2023). For example,

Fig. 3 Four pillars of food security



Nahoussé (2019) identified rainfall instability as a primary driver of inflation within the West African Economic and Monetary Union (WAEMU). Mukherjee and Ouattara (2021) also observed that temperature anomaly shocks contribute to inflationary pressures, with these effects persisting longer in developing poor countries than in wealthier nations (Iliyasu et al. 2023; Mukherjee and Ouattara 2021). Iliyasu et al. (2023) report that CC has increased food and consumer prices in South Africa, Nigeria, and Egypt, with a notably more significant impact on food prices than overall consumer price inflation (Iliyasu et al. 2023). Furthermore, Mawejje (2016) finds that temperature shocks account for more food price volatility in Uganda than rainfall (Mawejje 2016). Studies by Mawejje (2016) and Iliyasu et al. (2023) indicate that food prices, both domestically and internationally, are susceptible to the unpredictable impacts of CC (Iliyasu et al. 2023; Mawejje 2016). For example, previous research has demonstrated that CC positively affects global food prices. Iliyasu and Sanusi (2024) project that by 2050, global CC will increase rice prices by 32–37%, maize by 52–55%, wheat by 94–111%, and soybeans by 11–14% (Iliyasu and Sanusi 2024).

Increasing food prices may prompt individuals to opt for less expensive food options, potentially impacting their health. Evidence from Scotland indicates that during a recent rise in food costs, consumers shifted from more costly to cheaper food brands and avoided organic products (Parker 2018; Yang et al. 2014). Switching to less expensive food brands may have limited or beneficial nutritional effects, as some lower-cost brands feature reduced fat and sodium content, so reducing organic produce consumption is likely minimal (Varzakas and Smaoui 2024). Specific changes in food choices can be particularly concerning, as rising costs often lead individuals to select less healthy

options, given that more nutritious foods are more expensive than their less healthful counterparts (Bhardwaj et al. 2024). Energy-dense foods, typically cheaper than less energy-dense alternatives, are of particular concern. These foods are often highly processed and contain high sugar and fat levels. Due to the substantial portion of the cost attributed to processing and marketing, energy-dense foods are less susceptible to fluctuations in agricultural commodity prices. Consequently, Tchonkouang et al. (2024) found that major fast-food chains have experienced significant profit increases during recent food price surges despite an overall decline in consumer spending.

Consequently, the increase in food costs associated with CC may adversely affect the nutritional quality of diets and impair the nutritional status of specific populations. Rising food prices are also likely to elevate the risk of obesity, particularly among vulnerable groups such as children, young adults, smokers, lower-income populations, and elderly individuals who are frail and already have marginal nutritional status Scientific Advisory Committee on Nutrition (2012). The price rise is expected to worsen health disparities and pose equity challenges (Tchonkouang et al. 2024).

Adaptation in Food Production and Consumption

CC is affecting food production, consumption systems, and agricultural practices due to variations in regional climatic conditions, influencing adaptations to these changes. The food industry consists of production, processing, transportation, storage, preparation, marketing, and consumption, contributing about 15–30% of global GHG emissions, making it a significant source of GHG (Filho et al. 2022). According to Filho et al. (2022), the primary sources of GHG emissions within the food industry are agriculture (45%), food manufacturing (12%), and transportation (12%). The Agricultural Land Advisory Service (ADAS) suggests a strategy to mitigate GHG emissions is to feed cattle high-sugar grasses, which can reduce methane emissions. Another approach involves shifting the timing of animal manure application to land to decrease nitrous oxide emissions, another potent GHG. These modifications, however, may also have implications for food safety and nutritional quality (Filho et al. 2022). CC influences pesticide use, including fungicides and herbicides, by altering seasonal patterns and the prevalence of pests and diseases (Boxall et al. 2009). The effects will vary depending on the crop and geographic location. For example, Perrin et al. (2024) projected that global CC would increase pesticide use in the United States, with varying effects across different crops and regions scheduled pesticide application on wheat to increase by 14% in Kansas but decrease by 10% in Colorado (Perrin et al. 2024).

Similarly, pesticide use on maize in Illinois was anticipated to rise by 18%, whereas soybean usage was projected to increase by only 3%. Raising temperatures cause an increase in the introduction and re-emergence of diseases, vectors, or hosts, which could lead to a rise in the usage of biocides and veterinary drugs in livestock management of cattle (Perrin et al. 2024; Van Deynze et al. 2024). According to the Food and Agriculture Organization (FAO), this increase in veterinary interventions could contribute to a higher incidence of antibiotic-resistant diseases in human and animal populations.

Increasing water consumption for irrigation may be a critical agricultural adaptation strategy for CC. The global crop irrigation requirements could increase by 5–8%, with Southeast Asia experiencing rises of up to 15% (Srinivasan et al. 2024). However, the use of wastewater for irrigation, aimed at reducing water extraction, could elevate consumer exposure to pathogens. For instance, the 2008 outbreak of *Salmonella* serotype Saintpaul in the United States affected 1500 people and was linked to the production of irrigated wastewater from Mexico (Mishra et al. 2023). Additionally, increased irrigation may introduce contaminants such as pesticide residues and other chemicals into the food chain (Lebelo et al. 2021). As part of agricultural adaptation and CC mitigation efforts, new crop and livestock species will be developed or genetically modified to withstand diverse climatic conditions and produce fewer GHGs. Maintaining these new species' nutritional quality will require close monitoring (Soares et al. 2019).

Changes in food consumption patterns may affect food safety due to rising costs associated with CC, as different foods have varying levels of pesticide and chemical residues and various risks of foodborne illnesses. For example, increases in food prices may lead individuals to substitute more expensive fresh chicken with frozen chicken, increasing the risk of *Salmonella* contamination while reducing the risk of *Campylobacter* contamination (Kontovas and Psaraftis 2009). Predicting changes in food safety or nutritional outcomes is challenging without specific data on anticipated shifts in purchasing behaviors and dietary patterns (Habib et al. 2020).

Adaptation Strategies for Food Security Management

CC raises natural disasters and modifies crop productivity, ecosystem services, and water availability (Raza et al. 2019). These events result in fatalities and economic losses and hinder societal development. Adaptation and disaster risk management are crucial in mitigating these impacts while promoting social and economic resilience (Furtak and Wolińska 2023). The approaches necessitate the management of present and future risks. To guide adaptation strategies, it is essential to assess and prioritize threats, along with evaluating the capacity of communities to adapt and withstand these challenges (Biswas and Rahman 2023).

Developing nations depend highly on climate-sensitive sectors, such as agriculture and forestry, for food security and economic sustenance (Goli et al. 2024). Their vulnerability is exacerbated by a limited capacity to analyze climate risks and a lack of access to meteorological expertise (Raj et al. 2024). Deficiencies in infrastructure, communication networks, and resources further restrict their ability to implement timely disaster warnings. Effective adaptation and risk management strategies require comprehensive knowledge of exposure and vulnerability factors (Özekan and Akan 2023). Disaster management frameworks should integrate food security measures, including providing livestock shelters and seed storage facilities, along with robust systems for monitoring, evaluation, and early warning (Awuni et al. 2023).

CC and natural disasters significantly impact agricultural productivity and food security. Enhancements in seed quality, fertilizer usage, and water management

practices have the potential to reduce yield gaps (Usigbe et al. 2023). Knowledge-intensive agriculture equips farmers with advanced technologies and the support needed for adaptation. Adopting agronomic practices, infrastructure enhancements, disaster risk management tools, and developing improved crop varieties and agricultural technologies can contribute to economic growth and resilience (Khan 2024).

Information and communication technology (ICT) facilitates the timely delivery of critical data, empowering farmers and policymakers to make informed decisions (Kausar et al. 2023). Satellite technology allows for monitoring soil moisture and groundwater levels, enhancing water management, and optimizing crop scheduling (Kumar et al. 2023). Public access to satellite data is made available through online platforms and mobile applications. Additionally, various organizations offer hotline services to support agricultural practices in developing countries (Bhat et al. 2024).

Adaptation Strategies for Farm Practices

Smallholder farmers have employed various adaptive strategies to respond to environmental changes, including agroforestry, crop diversification, early planting, stress-tolerant crop varieties, the collection of wild edible plants, and mixed cropping techniques (Batungwanayo et al. 2023; Bhat et al. 2024). These practices enhance resilience to climate variability and contribute to improved agricultural sustainability. Crop diversification, in particular, is crucial in optimizing yields, enhancing soil fertility, increasing protein intake, and promoting dietary diversity within farming communities (Magesa et al. 2023). By cultivating various crops, farmers can better manage risks associated with pest outbreaks, fluctuating market prices, and adverse climatic conditions (Zulfikri et al. 2024). Selecting drought- and heat-resistant crop varieties is especially critical in regions prone to water scarcity and extreme temperatures. These resilient crop varieties can help stabilize and even boost agricultural production in challenging environments, ensuring food security in the face of climate stressors (Devate et al. 2023). Additionally, introducing new legume varieties, known for their nitrogen-fixing abilities, can further enhance soil fertility, reduce the need for synthetic fertilizers, and contribute to the overall resilience of farming systems. Together, these adaptive strategies support ecological sustainability and economic stability, enabling smallholder farmers to withstand the impacts of CC and better the effects of CC while maintaining productive and diverse agricultural landscapes (Adeagbo et al. 2023).

Agricultural technologies enhance crop productivity by optimizing soil and water management, utilizing alternative water resources, and expanding irrigation techniques (Adewusi et al. 2024). Drip irrigation minimizes environmental impact by delivering water and nutrients directly to the root zone, reducing waste. Essential farm practices include using high-efficiency crop varieties, drip irrigation systems, and lining waterways to minimize water loss (Yang et al. 2023). Additional methods, such as mulching, trenching, and applying organic fertilizers and pesticides, help crops better withstand climate stressors, improving overall productivity and resilience in agricultural systems (Abobatta and Fouad 2024). Enhancing the resilience of smallholder farmers in risk-

sensitive environments depends on participation in smaller-scale farming infrastructure. One of the tactics in the cattle industry is erecting animal shelters to guard against natural disasters and high temperatures. Regarding farming, water storage tanks and rooftop water-collecting systems guarantee the availability of required resources before or after a disaster (Babu et al. 2019).

Ecosystem restoration can aid in the adaptation to CC and disasters by enhancing diversity and variability. Institutional frameworks and stakeholder management plans are required for the restoration (Biswas and Rahman 2023). Crop improvement, soil conservation, conservation agriculture, forest conservation, integrated pest management, livestock, sustainable forest management, fisheries management, coral restoration, and fodder crops are all examples of adaptation strategies. These strategies conserve agricultural resources while lowering GHG emissions. Trees benefit communities by providing nutrients, feed, construction materials, and fuel. Food-insecure areas require systematic intervention to reverse soil deterioration and increase food production (Alotaibi 2023).

Climate shocks can disrupt food markets, humanitarian aid, government food reserves, and safety nets, exacerbating food insecurity. Strategic food reserves are critical in stabilizing supply and demand during crises, providing essential support to vulnerable populations, aiding communities and farmers, and maintaining fair market prices (Chitondo et al. 2024). Localized food stocks offer temporary relief in the event of delayed aid delivery. Farmers can safeguard seeds during emergencies by installing domestic seed storage facilities. Community seed banks further enhance resilience by preserving climate-adapted genetic diversity, contributing to the sustainability and effectiveness of these adaptive efforts (Olawale et al. 2023).

Capacity building enables individuals, groups, organizations, institutions, and entire societies to perform critical functions, address challenges, and achieve developmental objectives sustainably (Hashimy et al. 2023). Effective capacity building involves equipping skilled personnel and institutions with the ability to drive and oversee initiatives related to food security and other pressing needs. It is crucial for managing responses to natural disasters, CC, and fostering collaboration among diverse groups (Akintuyi 2024). Local capacity building is influenced by community resources, livelihoods, and assets, which shape its effectiveness and scope. Schools and student groups play a significant role in this process by participating in and managing disaster and climate risk mitigation strategies. By integrating educational institutions into capacity-building efforts, communities can enhance their preparedness and resilience, ensuring a more coordinated and practical approach to managing environmental and socio-economic challenges (Cid and Lerner 2023).

Prioritizing food security within institutional frameworks is crucial for effectively responding to natural disasters and CC. Strengthening governance systems, including national food security policies and legal frameworks, is essential for addressing these challenges (Zerbian and de Luis Romero 2023). Regional initiatives can promote sustainable agricultural practices and improve overall food security. Collaboration among policymakers, scientists, and professionals is necessary to develop and implement advanced decision support systems to better inform and guide responses to environmental and socio-economic stresses (Sithirith et al. 2024).

Global governments must establish protective measures and provide relocation support for displaced populations considering the climate migration crisis. By integrating comprehensive food security strategies and coordinated responses to migration, institutional frameworks can more effectively address the multifaceted impacts of CC and natural disasters, ensuring enhanced resilience and adaptive capacity for affected communities (Oñederra-Aramendi et al. 2023).

Recommendation

RQ-1. The outcomes of integrating multiple adaptation strategies on food security, recognizing the use of a combination of approaches:

Based on the studies, the integration of multiple adaptation strategies can improve food security by mitigating the effects of CC. Adaptation strategies in food production include optimizing seed quality, fertilizer usage, and water management practices to enhance agricultural productivity and reduce yield gaps (Usigbe et al. 2023). Resilient crop varieties, improved agricultural technologies, and agronomic practices contribute to economic resilience and sustainability (Khan 2024). Additionally, diversification in crop varieties, agroforestry, and stress-tolerant crops enhance soil fertility and optimize yields, improving dietary diversity and managing risks associated with pest outbreaks and climate variability (Magesa et al. 2023). Smallholder farmers can build resilience to climate stressors by employing diverse adaptive strategies and maintaining sustainable, productive agricultural landscapes (Adeagbo et al. 2023).

RQ-2. How adaptive agricultural practices can improve food security under the impacts of CC:

Adaptive agricultural practices enhance food security by addressing CC impacts on agricultural ecosystems. For example, developing drought- and heat-resistant crop varieties helps stabilize agricultural production in regions prone to water scarcity and extreme temperatures, ensuring food security in challenging environments. Agriculture, sustainable land use practices, and integrated pest management further strengthen the ecosystem's ability to sustain food production. Adapting water management approaches to mitigate the impacts of climate change. Water use and integrated management are vital but often fall short due to barriers such as inadequate infrastructure, fragmented governance, and limited stakeholder engagement. (Devate et al. 2023). Enhancing water use efficiency, particularly in agriculture, through advanced technologies like precision irrigation and promoting integrated water resources management, drip irrigation, mulching, and trenching help optimize water management and minimize environmental impacts, and increasing crop resilience to climate stressors are crucial steps forward. (Abobatta and Fouad 2024). Adaptive management practices that allow flexibility in response to CC conditions must be more widely implemented. Continued innovation, strengthened governance,

and cross-sectoral collaboration are essential to effectively enhance support of sustainable food production in adverse climatic conditions, improve overall agricultural productivity, and reduce the risks of water scarcity and droughts in the face of ongoing climate challenges.(Adewusi et al. 2024).

RQ-3. Non-farm options that support the sustainability of livelihoods when agricultural activities face threats from CC:

Non-farm adaptation strategies in supporting livelihoods amidst agricultural threats due to CC. For instance, ecological restoration efforts, such as agro-processing, sustainable forest management, coral restoration, and integrated pest management, help conserve agricultural resources while lowering GHGs (Babu et al. 2019). Livestock adaptation strategies, such as building animal shelters to protect against natural disasters, contribute to livelihood sustainability (Babu et al. 2019). Information and communication technologies (ICTs) also aid livelihood sustainability by delivering critical data to farmers and policymakers, enabling informed agricultural and economic planning (Kausar et al. 2023).

RQ-4. The implications of various CCA strategies on food security and suggestions for future research in this area:

CCA strategies have broad implications for food security, such as enhancing resilience to climate-related disasters and optimizing agricultural productivity (Kausar et al. 2023). Early warning systems, evaluation, and monitoring frameworks are crucial for disaster risk management in developing countries, which rely heavily on climate-sensitive sectors like agriculture and forestry (Semeraro et al. 2023). As CC continues to alter global hydrological cycles and disrupt food production, future research should focus on developing region-specific adaptation strategies, improving access to climate risk analysis, and enhancing technologies that facilitate sustainable food production in vulnerable areas (Furtak and Wolińska 2023; Semeraro et al. 2023). There is also a need to investigate the long-term impacts of genetic crop modifications and improved irrigation techniques on food safety, nutrition, and environmental sustainability (Fan et al. 2008).

Conclusion

The impacts of CC on global food security are profound, challenging the stability and sustainability of food systems worldwide. As climate extremes become more frequent, they place immense pressure on agricultural production and distribution, increasing food insecurity and volatility in global food prices. This review underscores the need for a multifaceted approach to CC adaptation, emphasizing the integration of sustainable agricultural practices, such as crop diversification, resilient crop varieties, and improved water management techniques. Non-farm alternatives

and robust food reserve systems also emerge as crucial components in maintaining food security, especially during times of crisis.

Furthermore, collaborative efforts between policymakers, researchers, and practitioners are imperative to ensure these adaptation strategies are effective and scalable. Strengthening governance systems and advancing capacity-building initiatives will help local communities and nations better manage climate-related risks. Investment in innovative technologies will be crucial in enhancing adaptive capacity and improving resilience within food systems.

Future research should aim to develop holistic solutions that integrate multiple adaptation strategies, exploring their potential synergies to provide a more comprehensive and effective response to climate-induced challenges. By prioritizing these strategies, we can protect global food systems, ensuring a secure and sustainable food future for all.

Competing Interest Declaration The author(s) has no competing interests to declare that are relevant to the content of this manuscript.

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